Constrained Deep Reinforcement Learning with Lagrangian Theory

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Abstract

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Contents

1 Introduction **1**
1.1 Background 1
1.2 Research Objectives 1
2 Notations and Preliminaries **1**
3 Review of Contraction Theory **2**
3.1 Basic Results of Contraction Theory for Deterministic Systems 2
3.1.1 Notable Definitions 2
3.1.2 Fundamental Theorem of Contraction Theory 3
3.1.3 Partial Contraction 4
3.2 Basic Results of Contraction Theory for Stochastic Systems 4
4 Conclusion **4**

1 Introduction

1.1 Background

1.2 Research Objectives

The main objectives of this research are as follows:

- Mathematical stability analysis of the controller and estimator with deep neural networks using the contraction theory.
- Development of the controller and estimator with deep neural networks using the contraction theory.

2 Notations and Preliminaries

The following notations are used throughout this document:

- $:=$ denotes *defined as*.
- $(\cdot)^\top$ denotes the transpose of a matrix or a vector.
- $\boldsymbol{x} := [x_i]_{i\in\{1,\cdots,n\}} \in \mathbb{R}^n$ denotes the state vector.
- $\boldsymbol{A} := [a_{ij}]_{i,j\in\{1,\cdots,n\}} \in \mathbb{R}^{n\times n}$ denotes a matrix.
- $\lambda_i(\boldsymbol{A}),\ i \in \{\max, \min\}$ denotes the maximum and minimum singular value of $\boldsymbol{A}$, respectively.

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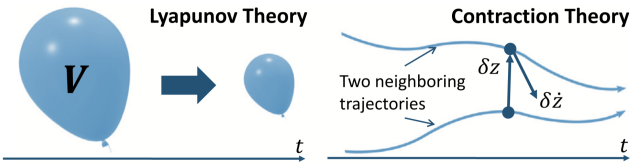

Figure 1: Difference between Lyapunov and contraction theory [4, Fig. 1]. The Lyapunov theory investigates the convergence to a single point and the contraction theory does regarding a single trajectory.

- $\boldsymbol{I}_n$ denotes the identity matrix of size n and $\boldsymbol{0}_{n\times m}$ denotes the zero matrix of size $n \times m$.
- sym denotes the symmetric part of a matrix, i.e., $\mathrm{sym}(\boldsymbol{A}) := \boldsymbol{A} + \boldsymbol{A}^\top$ [1].

We introduce the following lemmas.

Lemma 1 (Comparison Lemma)**.** *Suppose that a continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ satisfies the following inequality:*

$$\tfrac{\mathrm{d}}{\mathrm{d}t} f(t) \leq -af(t) + b, \quad \forall t \in \mathbb{R}_{\geq 0},$$

where $a, b > 0$. Then, the following inequality holds:

$$f(t) \leq -af(0)e^{-at} + \tfrac{b}{a}(1 - e^{-at}), \quad \forall t \in \mathbb{R}_{\geq 0}$$

and remains in a compact set $f(t) \in \{\|f(t)\| \mid \|f(0)\| \leq \frac{b}{a}\}$.

Proof. This is a simple special case of the comparison lemma [2, pp. 102-103]. See [2, pp. 659-660]. □

3 Review of Contraction Theory

For your smooth start, we recommend you to begin with [3]. The overview of contraction theory is presented in a review paper [4].

3.1 Basic Results of Contraction Theory for Deterministic Systems

First, we start with the following deterministic systems:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t), \tag{1}$$

where $\boldsymbol{f}(\boldsymbol{x}, t)$ is an $n \times 1$ sufficiently smooth non-linear vector function and $\boldsymbol{x} \in \mathbb{R}^n$ is the state vector. The smooth property of $\boldsymbol{f}(\boldsymbol{x}, t)$ is essential to ensure the existence and uniqueness of the solution to (1) [2, see, pp. 88-89].

The biggest difference between the traditional Lyapunov theory and the contraction theory is that the contraction theory investigates the convergence of the state trajectory to a single trajectory (contraction behavior), while the Lyapunov theory focuses on the convergence of the state trajectory to a single point i.e., see, Fig. 1. For this, motivated by the calculus of variations [5, Chap. 4], (1) can be rewritten as differential dynamics using *differential displacement* $\delta\boldsymbol{x}$ as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}(\boldsymbol{x}, t)\delta\boldsymbol{x}. \tag{2}$$

For your information, $\delta\boldsymbol{x}$ is an infinitesimal displacement at *fixed time*.

3.1.1 Notable Definitions

Before we present the fundamental theorem of contraction theory, we introduce the following definitions. One can re-visit this section while reading further.

Definition 1 (see, Def. 2.2 [4])**.** If any two trajectories $\boldsymbol{\xi}_1(t)$ and $\boldsymbol{\xi}_2(t)$ of (1) converge to a single trajectory, then the system (1) is said to be *incrementally exponentially stable*, if $\exists C, \alpha > 0$, subject to the following holds:

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| \leq C\|\boldsymbol{\xi}_1(0) - \boldsymbol{\xi}_2(0)\| \exp^{-\alpha t}, \ \forall t \in \mathbb{R}_{\geq 0}.$$

The result of Theorem 1 equivalently implies the incremental exponential stability, since we have $\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| = \|\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \delta\boldsymbol{x}(t)\|$.

Definition 2. Let $\boldsymbol{\Theta}(\boldsymbol{x},t)$ be a smooth coordinate transformation of $\delta\boldsymbol{x}$ to $\delta\boldsymbol{z}$, i.e., $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$. Then, a symmetric continuously differentiable matrix $\boldsymbol{M}(\boldsymbol{x},t) := \boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t)$ is said to be a *metric* of the system (1).

Definition 3. The covariant derivative of $\boldsymbol{f}(\boldsymbol{x},t)$ in $\delta\boldsymbol{x}$ coordinate is represented as

$$\boldsymbol{F} := \left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} + \boldsymbol{\Theta}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)\boldsymbol{\Theta}^{-1},$$

and is called the *generalized Jacobian*. This can be easily derived by differentiating $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$ with respect to t, leading to $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{z} = \boldsymbol{F}\boldsymbol{z}$.

3.1.2 Fundamental Theorem of Contraction Theory

The following theorem presents the fundamental theorem of contraction theory and corresponding necessary and sufficient condition for exponential convergence of the differential system (2).

Theorem 1 (see, T. 2.1 [4])**.** *If $\exists \boldsymbol{M}(\boldsymbol{x},t) = \boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t) > 0, \forall \boldsymbol{x},t$ where $\boldsymbol{\Theta}(\boldsymbol{x},t)$ defines a smooth coordinate transformation of $\delta\boldsymbol{x}$ to $\delta\boldsymbol{z}$, i.e., $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$, subject to the following equivalent conditions holds for $\exists\alpha \in \mathbb{R}_{>0},\ \forall \boldsymbol{x},t$:*

$$\lambda_{\max}(\boldsymbol{F}(\boldsymbol{x},t)) = \lambda_{\max}\left(\left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} + \boldsymbol{\Theta}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)\boldsymbol{\Theta}^{-1}\right) \leq -\alpha,$$
$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + \mathrm{sym}(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}) \leq -2\alpha\boldsymbol{M},$$

where the arguments of $\boldsymbol{M}(\boldsymbol{x},t)$ and $\boldsymbol{F}(\boldsymbol{x},t)$ are omitted for simplicity, then, the system (1) is said to be contracting with an exponential rate α, i.e., all trajectories of (1) converge to a single trajectory. The converse is also true.

Proof. Taking time derivative of a differential Lyapunov function of $\delta\boldsymbol{x}$, $V = \delta\boldsymbol{z}^\top\boldsymbol{z} = \delta\boldsymbol{x}^\top\boldsymbol{M}\delta\boldsymbol{x}$, using the differential dynamics (2), we have

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}V(\boldsymbol{x},\delta\boldsymbol{x},t) =& \delta\boldsymbol{x}^\top\left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + \mathrm{sym}(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}})\right)\delta\boldsymbol{x} = 2\delta\boldsymbol{z}^\top\boldsymbol{F}\delta\boldsymbol{z}\\ \leq& -2\alpha\delta\boldsymbol{x}^\top\boldsymbol{M}\delta\boldsymbol{x} = -2\alpha\delta\boldsymbol{z}^\top\delta\boldsymbol{z} = -2\alpha V.\end{aligned}$$

According to the comparison lemma (Lemma 1), we have $V(t) \leq V(0)e^{-2\alpha t}$, which then yields $\|\delta\boldsymbol{z}(t)\|^2 \leq \|\delta\boldsymbol{z}(0)\|^2 e^{-2\alpha t}$ and $\|\delta\boldsymbol{z}(t)\| \leq \|\delta\boldsymbol{z}(0)\|e^{-\alpha t}$. This implies that any infinitesimal displacement $\delta\boldsymbol{z}$ and $\delta\boldsymbol{x}$ converges to zero exponentially with rate α. Note that the initial conditions are exponentially "forgotten" as time goes on. The proof of the converse can be found in [3, Sec. 3.5]. □

It is notable that the unboundedness of the metric $\boldsymbol{M}(\boldsymbol{x},t)$ does not create any problem in a technical sense. This is because, the dynamics of $\boldsymbol{M}(\boldsymbol{x},t)$ is linear with infinite escape time. Therefore, it can be handled by renormalizing the metric $\boldsymbol{M}(\boldsymbol{x},t)$.

Theorem 1 can also be proven by using the transformed squared length integrated over two arbitrary solutions of (1). The following theorem presents the alternative proof of Theorem 1.

Theorem 2 (see, T. 2.3 [4])**.** *Let $\boldsymbol{\xi}_1(t)$ and $\boldsymbol{\xi}_2(t)$ be two solutions of (1), and define the transformed squared length with $\boldsymbol{M}(\boldsymbol{x},t)$ of Theorem 1 as follows:*

$$V_{sl}(\boldsymbol{x},\delta\boldsymbol{x},t) = \textstyle\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)}\|\delta\boldsymbol{z}\|^2 = \int_0^1 \frac{\partial\boldsymbol{x}}{\partial\mu}^\top\boldsymbol{M}(\boldsymbol{x},t)\frac{\partial\boldsymbol{x}}{\partial\mu}\mathrm{d}\mu, \tag{4}$$

where $\boldsymbol{x}$ is a smooth path parameterized as $\boldsymbol{x}(\mu = 0,t) = \boldsymbol{\xi}_1(t)$ and $\boldsymbol{x}(\mu = 1,t) = \boldsymbol{\xi}_2(t)$ by $\mu \in \{0,1\}$. Also, define the path integral with the transformation $\boldsymbol{\Theta}(\boldsymbol{x},t)$ for $\boldsymbol{M}(\boldsymbol{x},t) = \boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t)$ as follows:

$$V_l(\boldsymbol{x},\delta\boldsymbol{x},t) = \textstyle\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)}\|\delta\boldsymbol{z}\| = \int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)}\|\boldsymbol{\Theta}(\boldsymbol{x},t)\delta\boldsymbol{x}\|. \tag{5}$$

Then, (4) and (5) satisfy the following inequality:

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| = \|\textstyle\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)}\delta\boldsymbol{x}\| \leq \frac{V_l}{\sqrt{\underline{m}}} \leq \sqrt{\frac{V_{sl}}{\underline{m}}},$$

where $\boldsymbol{M}(\boldsymbol{x},t) \geq \underline{m}\boldsymbol{I}_n,\ \forall \boldsymbol{x},t$ for $\exists\underline{m} \in \mathbb{R}_{>0}$ and Theorem 1 can also be proven by using (4) and (5) as a Lyapunov-like function, resulting in incremental exponential stability of the system (1) (see, Definition 1). Note that the shortest path integral V_l of (5) with a parameterized state $\boldsymbol{x}$ (i.e., $\inf V_l = \sqrt{\inf V_{sl}}$) defines the Riemannian distance and the path integral of a minimizing geodesic.

Proof. ... □

Example 1. Consider the following system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & x_1 \\ -x_1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

However, aforementioned theorems are limited to the convergence of the state trajectory to a single trajectory. In the next section, we introduce the partial contraction theory to investigate the convergence of the state trajectory to a desired trajectory.

3.1.3 Partial Contraction

[6, 7]

3.2 Basic Results of Contraction Theory for Stochastic Systems

4 Conclusion

References

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